A NONZERODIVISOR IN A ONE-DIMENSIONAL NOETHERIAN LOCAL RING IS ONLY BOUNDEDLY DIVISIBLE BY A PARAMETER IN FINITE EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. All three referenced results live in earlier chapters, so each locator points at the verbatim line of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carries the canonical tag.

Allowed standard prerequisites (editorial).

- Dominating a one-dimensional Noetherian domain by a discrete valuation ring. Algebra, Lemma labelled algebra-lemma-dominate-by-dimension-1 (canonical Stacks tag 00P8); invoked at proof lines 615-617 and again at proof lines 633-634. Supplies the discrete valuation ring A dominating R/q with the same fraction field, and later the ring A"' dominating A"_m"; the valuation estimate is the counted content.
- Surjectivity of spectra for integral and for radical extensions. Algebra, Lemma algebra-lemma-integral-overring-surjective (00GQ, proof lines 625-626) and Lemma algebra-lemma-surjective-spec-radical-ideal (00FI, proof lines 627-628). Supply that Spec(R') -> Spec(R) and Spec(A') -> Spec(A) are surjective, which is what produces the prime q' of A' lying over (0) of A.

Source reference. Original lemma 00P8: sources/stacks-chow/tags.txt, source lines 922–922. Cross-chapter standard prerequisite invoked at proof lines 615-617 and again at 633-634 to obtain the two discrete valuation rings; the chain of base changes and the valuation comparison are retained in the counted span.

Source reference. Original lemma 00GQ: sources/stacks-chow/tags.txt, source lines 616–616. Cross-chapter standard prerequisite invoked once at proof lines 625-626 for surjectivity of Spec(R') -> Spec(R).

Source reference. Original lemma 00FI: sources/stacks-chow/tags.txt, source lines 572–572. Cross-chapter standard prerequisite invoked once at proof lines 627-628 for surjectivity of Spec(A') -> Spec(A); the choice of the prime q' and the rest of the argument are retained in the counted span.

Date: stacks-project:chow.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $a, b \in R$ be nonzerodivisors with $a \in \mathfrak{m}$. There exists an integer $n = n(R, a, b)$ such that for a finite ring extension $R \subset R'$ if $b = a^m c$ for some $c \in R'$, then $m \leq n$.*

Proof. Choose a minimal prime $\mathfrak{q} \subset R$. Observe that $\dim(R/\mathfrak{q}) = 1$, in particular $R/\mathfrak{q}$ is not a field. We can choose a discrete valuation ring A dominating $R/\mathfrak{q}$ with the same fraction field, see Algebra, Lemma 00P8. Observe that a and b map to nonzero elements of A as nonzerodivisors in R are not contained in $\mathfrak{q}$. Let v be the discrete valuation on A. Then $v(a) > 0$ as $a \in \mathfrak{m}$. We claim $n = v(b)/v(a)$ works.

Let $R \subset R'$ be given. Set $A' = A \otimes_R R'$. Since $\mathrm{Spec}(R') \to \mathrm{Spec}(R)$ is surjective (Algebra, Lemma 00GQ) also $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is surjective (Algebra, Lemma 00FI). Pick a prime $\mathfrak{q}' \subset A'$ lying over $(0) \subset A$. Then $A \subset A'' = A'/\mathfrak{q}'$ is a finite extension of rings (again inducing a surjection on spectra). Pick a maximal ideal $\mathfrak{m}'' \subset A''$ lying over the maximal ideal of A and a discrete valuation ring A''' dominating $A''_{\mathfrak{m}''}$ (see lemma cited above). Then $A \to A'''$ is an extension of discrete valuation rings and we have $b = a^m c$ in A'''. Thus $v'''(b) \geq mv'''(a)$. Since $v''' = ev$ where e is the ramification index of A'''/A, we find that $m \leq n$ as desired. $\square$